

Positive Families for the Remaining One-Sided James Question

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Status and Source Problem

This packet concerns the final remarks of Cascales–Orihuela–Pérez, *One-side James' Compactness Theorem* [1]. The relevant passage asks whether their Theorems 2 and 10 remain valid for arbitrary Banach spaces:

boundness of $\tilde{f}^{-1}((-\infty, \mu])$ and the proof is over. □

In arbitrary Banach spaces the previous theorem is not true, since the weak*-version of James' theorem for a separable Banach space E holds if and only if E does not contain $\ell^1(\mathbb{N})$, see Corollary II.13 in [10].

6. FINAL REMARKS AND QUESTIONS

- (i) It is not possible to extend Theorem 10 for more general sets $D \subset E$. To see this, suppose that D satisfies the property that *whenever B is a bounded subset of E such that every $x^* \in L_D(E^*, E)$ attains its supremum on B , then B is weakly relatively compact*. In particular, the set $B = \overline{\text{co}(\{0\} \cup D)}^{\|\cdot\|}$ is weakly compact, since every $x^* \in L_D(E^*, E)$ attains its supremum on B at 0. But this implies in particular that D is weakly relatively compact.
- (ii) We do not know if Theorems 2 and 10 are valid for arbitrary Banach spaces. We conjecture that the answer is affirmative in both cases.
- (iii) Let E be a Banach space. If E does not contain an isomorphic copy of ℓ^1 then (B_{E^*}, ω^*) is convex block compact (see [1, Prop. 3.11]). This result was strengthened by Pfitzner [19, p. 601, Proposition 11], who proved that if E does not contain asymptotically isometric copies of ℓ^1 , then (B_{E^*}, ω^*) is convex block compact.

On the other hand, it can be shown that if (B_{E^*}, ω^*) is block compact then E does not contain a copy of $\ell^1(\mathbb{R})$ (see [16] for the details). Haydon, Levy and Odell [11] proved that under Martin Axiom plus the negation of the Continuum Hypothesis, if E does not contain a copy of $\ell^1(\mathbb{R})$ then (B_{E^*}, ω^*) is convex block compact. In particular, this implies that convex block compactness and block compactness on (B_{E^*}, ω^*) co-

Orihuela's later paper *Conic James' Compactness Theorem* [2] explicitly resolves the Theorem 10 part for arbitrary Banach spaces and states that Theorem 2 remains unknown. The

present packet consolidates a stronger partial analysis of that remaining Theorem 2: an exact one-set reduction, an affine-slice theorem, a conic-dominated family obtained from [2], and a direct frustum-family theorem and finite-dimensional cylinder theorem showing that broad attempted-counterexample classes actually force weak compactness.

The Remaining Two-Set Question

Theorem 2 of [1] says, under the additional hypothesis that (B_{E^*}, w^*) is convex block compact, that if A and B are bounded closed convex subsets of a Banach space E with $\text{dist}(A, B) > 0$, and if every functional $x^* \in E^*$ satisfying

$$\sup(x^*, B) < \inf(x^*, A)$$

attains its supremum on B and its infimum on A , then both A and B are weakly compact. The open part is whether the convex-block-compactness assumption on the dual ball can be removed.

Theorem 1 (Exact one-set reduction). *The arbitrary-Banach-space version of the preceding two-set statement is equivalent to the following one-set assertion:*

If $C \subset E$ is nonempty, bounded, closed, and convex, with $\text{dist}(0, C) > 0$, and every $x^ \in E^*$ satisfying*

$$\sup_C x^* < 0$$

attains its supremum on C , then C is weakly compact.

Proof. The two-set statement implies the one-set assertion by taking $A = \{0\}$ and $B = C$. Indeed, $\inf(x^*, A) = 0$, so the two-set attainment hypothesis is exactly the one-sided hypothesis on C , while attainment on A is automatic.

Conversely, assume the one-set assertion and suppose $A, B \subset E$ satisfy the two-set hypotheses. Put

$$C = \overline{B - A}^{\|\cdot\|}.$$

Then C is nonempty, bounded, closed, and convex, and $\text{dist}(0, C) = \text{dist}(A, B) > 0$. If $\sup_C x^* < 0$, then

$$\sup_{b \in B} x^*(b) < \inf_{a \in A} x^*(a).$$

By the two-set hypothesis there are $b_0 \in B$ and $a_0 \in A$ at which x^* attains the corresponding supremum and infimum. Hence

$$x^*(b_0 - a_0) = \sup_{b \in B} x^*(b) - \inf_{a \in A} x^*(a) = \sup_C x^*,$$

so x^* attains its supremum on C . The one-set assertion gives weak compactness of C . Finally, fixing $b_1 \in B$ and $a_1 \in A$, we have $A \subset b_1 - C$ and $B \subset a_1 + C$. Since A and B are norm-closed convex sets, they are weakly closed; therefore they are weakly compact subsets of weakly compact translates of C . $\square$

Idea of the Proof

Theorem 1 isolates the hard core: prove weak compactness of C from attainment only on the strict negative cone $\{x^* : \sup_C x^* < 0\}$. In the affine-slice case there is a simple way to do this. If a functional φ is constant and negative on C , then for any given $y^* \in E^*$ we can choose $\lambda > 0$ so large that $y^* + \lambda\varphi$ is strictly negative on C . This shifted functional attains its supremum by hypothesis, and since φ is constant on C , the same point also maximizes y^* .

The frustum theorem below attacks a different possible source of counterexamples. If C is the convex hull of a point in one coordinate slice and a bounded closed convex base S in a farther parallel slice, then the negative cone can be tuned to expose the base. Thus the one-sided hypothesis forces every functional on the base space to attain on S , and James' theorem makes S weakly compact.

The finite-dimensional cylinder theorem uses a still simpler localization. If $C = P \times S$ and P is a finite-dimensional compact convex set separated from 0, a large negative separator on P pushes any prescribed $y^* \in Y^*$ into the strict negative cone without disturbing the S -coordinate. Attainment on C then forces y^* to attain on S .

Lemma 1 (One-set affine-slice form). *Let E be a Banach space, and let $C \subset E$ be nonempty, bounded, closed, and convex. Assume that $\text{dist}(0, C) > 0$ and that every $x^* \in E^*$ with $\sup_C x^* < 0$ attains its supremum on C . If there is $\varphi \in E^*$ and $\alpha < 0$ such that*

$$\varphi(c) = \alpha \quad \text{for every } c \in C,$$

then C is weakly compact.

Proof. Let $y^* \in E^*$ be arbitrary. Since C is bounded, the number $M = \sup_C y^*$ is finite. Choose $\lambda > 0$ so large that

$$M + \lambda\alpha < 0.$$

Then

$$\sup_C (y^* + \lambda\varphi) = \sup_C y^* + \lambda\alpha < 0.$$

By hypothesis, $y^* + \lambda\varphi$ attains its supremum on C , say at $c_0 \in C$. Since φ is constant on C , for every $c \in C$ we have

$$y^*(c) + \lambda\alpha = (y^* + \lambda\varphi)(c) \leq (y^* + \lambda\varphi)(c_0) = y^*(c_0) + \lambda\alpha.$$

Thus y^* attains its supremum on C . As y^* was arbitrary, every element of E^* attains its supremum on C . James' compactness theorem now implies that C is weakly compact. $\square$

Theorem 2 (Affine-slice subcase of the remaining one-sided James theorem). *Let E be a Banach space, and let $A, B \subset E$ be nonempty bounded closed convex sets with $\text{dist}(A, B) > 0$. Assume that:*

1. *every $x^* \in E^*$ with*

$$\sup(x^*, B) < \inf(x^*, A)$$

attains its supremum on B and its infimum on A ;

2. *there are $\varphi \in E^*$ and real numbers $\gamma < \delta$ such that*

$$\varphi(b) = \gamma \quad (b \in B), \quad \varphi(a) = \delta \quad (a \in A).$$

Then A and B are weakly compact.

Proof. Put $C = \overline{B - A}^{\|\cdot\|}$. Then C is nonempty, bounded, closed, and convex. Since A and B are norm-closed convex sets and $\text{dist}(A, B) > 0$, we have $0 \notin C$, indeed $\text{dist}(0, C) = \text{dist}(A, B) > 0$.

Let $x^* \in E^*$ satisfy $\sup_C x^* < 0$. Since

$$\sup_C x^* = \sup_{b \in B} x^*(b) - \inf_{a \in A} x^*(a),$$

we have

$$\sup(x^*, B) < \inf(x^*, A).$$

By hypothesis, there are $b_0 \in B$ and $a_0 \in A$ such that

$$x^*(b_0) = \sup(x^*, B), \quad x^*(a_0) = \inf(x^*, A).$$

Therefore $b_0 - a_0 \in C$ and

$$x^*(b_0 - a_0) = \sup_{b \in B} x^*(b) - \inf_{a \in A} x^*(a) = \sup_C x^*.$$

Thus every functional strictly negative on C attains its supremum on C .

The affine-slice assumption gives

$$\varphi(b - a) = \gamma - \delta < 0 \quad (b \in B, a \in A).$$

By norm-continuity, $\varphi(c) = \gamma - \delta$ for every $c \in C$. Lemma 1 applies and shows that C is weakly compact.

Fix $b_1 \in B$ and $a_1 \in A$. For every $a \in A$,

$$a = b_1 - (b_1 - a) \in b_1 - C,$$

so $A \subset b_1 - C$. Since A is norm-closed and convex, it is weakly closed; hence A is weakly compact as a weakly closed subset of the weakly compact set $b_1 - C$. Similarly,

$$b = a_1 + (b - a_1) \in a_1 + C$$

for every $b \in B$, so B is weakly compact. □

A Conic-Dominated Family

The supporting paper [2] proves the following result for arbitrary Banach spaces: if $D \subset E$ is weakly compact with $0 \notin D$, and every $x^* \in E^*$ which is positive on D attains its supremum on a bounded set A , then A is weakly relatively compact. This extends Theorem 10 of [1]; the paper explicitly notes that Theorem 2 of [1] remains unknown for arbitrary Banach spaces.

This theorem gives another positive family for the one-set reduction.

Theorem 3 (Conic-dominated one-sided sets). *Let $C \subset E$ be nonempty, bounded, closed, and convex, with $\text{dist}(0, C) > 0$. Assume every $x^* \in E^*$ satisfying $\sup_C x^* < 0$ attains its supremum on C . Suppose moreover that there is a weakly compact set $D \subset E$, with $0 \notin D$, such that*

$$x^*(D) > 0 \implies \sup_C x^* < 0,$$

where $x^(D) > 0$ is the notation of [2] for positivity on the whole direction set D . Then C is weakly compact.*

Proof. By the displayed implication and the one-sided hypothesis, every $x^* \in E^*$ with $x^*(D) > 0$ attains its supremum on C . Orihuela's conic James theorem [2] therefore implies that C is weakly relatively compact. Since C is norm-closed and convex, it is weakly closed, and hence weakly compact. □

A Frustum Family

The next theorem records a direct family check which grew out of trying to construct counterexamples. It shows that cone-over-a-base examples are too visible to the strict negative cone: if the base is not weakly compact, James' theorem supplies a functional on the base which can be tuned into a nonattaining negative separator.

Theorem 4 (One-dimensional frustum family). *Let Y be a Banach space, let $S \subset Y$ be nonempty, bounded, closed, and convex, and assume $0 \in S$. Fix $0 < \alpha < \beta$, put $E = \mathbb{R} \oplus_\infty Y$, and define*

$$C = \text{co}(\{(\alpha, 0)\} \cup (\{\beta\} \times S)) \subset E.$$

If every $x^ \in E^*$ satisfying $\sup_C x^* < 0$ attains its supremum on C , then S is weakly compact. Consequently C is weakly compact.*

Proof. Every functional on $E = \mathbb{R} \oplus_\infty Y$ has the form

$$f_{t,y^*}(s, y) = -ts + y^*(y)$$

with $t \in \mathbb{R}$ and $y^* \in Y^*$. Fix $y^* \in Y^*$ and write $h = \sup_S y^*$. Since $0 \in S$, $h \geq 0$. If $h = 0$, then y^* attains its supremum on S at 0.

Assume now that $h > 0$. Choose $t > 0$ such that

$$\frac{h}{\beta} < t < \frac{h}{\beta - \alpha};$$

this is possible because $0 < \alpha < \beta$. For $f = f_{t,y^*}$, the value at the apex $(\alpha, 0)$ is $-t\alpha$, while the supremum over the far slice $\{\beta\} \times S$ is $-t\beta + h$. The choice of t gives

$$-t\beta + h < 0 \quad \text{and} \quad -t\beta + h > -t\alpha.$$

Since a linear functional has the same supremum on the convex hull as on the generating set,

$$\sup_C f = -t\beta + h < 0.$$

By the one-sided hypothesis, f attains its supremum on C . The strict inequality $-t\beta + h > -t\alpha$ forces every maximizing point to lie in the far slice $\{\beta\} \times S$. Therefore there is $y_0 \in S$ such that

$$-t\beta + y^*(y_0) = -t\beta + h,$$

and so $y^*(y_0) = h$. Thus every $y^* \in Y^*$ attains its supremum on S . By James' compactness theorem, S is weakly compact. Finally, C is the image of the weakly compact product $[0, 1] \times S$ under the continuous affine map

$$(\lambda, y) \mapsto \lambda(\alpha, 0) + (1 - \lambda)(\beta, y),$$

so C is weakly compact. □

A Finite-Dimensional Cylinder Family

The following theorem is useful because it covers non-affine-slice sets: the finite-dimensional factor P may have many points. It says that a non-weakly-compact cylinder base cannot be hidden from the one-sided negative cone.

Theorem 5 (Finite-dimensional cylinders). *Let F be a finite-dimensional Banach space and let Y be a Banach space. Let $P \subset F$ be nonempty, compact, and convex with $\text{dist}(0, P) > 0$, and let $S \subset Y$ be nonempty, bounded, closed, and convex. Put*

$$E = F \oplus_\infty Y, \quad C = P \times S \subset E.$$

If every $x^ \in E^*$ satisfying $\sup_C x^* < 0$ attains its supremum on C , then S is weakly compact. Consequently C is weakly compact.*

Proof. Since P is compact, convex, and separated from 0, the separation theorem gives $\varphi \in F^*$ such that

$$m := \sup_P \varphi < 0.$$

Let $y^* \in Y^*$ be arbitrary and write $M = \sup_S y^*$. Pick $\lambda > 0$ so large that $\lambda m + M < 0$, and define

$$x^*(p, y) = \lambda \varphi(p) + y^*(y) \quad (p \in P, y \in S).$$

Then

$$\sup_C x^* = \lambda \sup_P \varphi + \sup_S y^* = \lambda m + M < 0.$$

By the one-sided hypothesis, x^* attains its supremum on C , say at $(p_0, y_0) \in P \times S$. Holding p_0 fixed, for every $y \in S$ we have

$$\lambda \varphi(p_0) + y^*(y) = x^*(p_0, y) \leq x^*(p_0, y_0) = \lambda \varphi(p_0) + y^*(y_0).$$

Thus y^* attains its supremum on S . Since y^* was arbitrary, James' compactness theorem implies that S is weakly compact. The product $P \times S$ is then weakly compact, since P is compact finite-dimensional and S is weakly compact. $\square$

Limitations and Novelty Check

This is not a full solution of the remaining conjecture. The general one-set reduction asks whether a bounded closed convex set C with $\text{dist}(0, C) > 0$ must be weakly compact whenever every x^* with $\sup_C x^* < 0$ attains its supremum on C . The affine-slice proof works only when there is a continuous functional constant and negative on C . For a general separating functional, adding a large multiple of it can change the maximizing behavior, so the same shift argument does not settle the problem. The conic-dominated theorem uses [2] and only applies when the negative cone contains the positive cone of a fixed weakly compact direction set. The frustum and cylinder theorems rule out two broad elementary counterexample shapes, but they do not cover arbitrary convex bodies away from 0.

Equivalently, the remaining hard problem is to show, or refute, the following localization of James' theorem: whenever such a set C is not weakly compact, there is a functional x^* in the strict negative cone $\{x^* : \sup_C x^* < 0\}$ which does not attain its supremum on C .

I checked the local run indexes for arXiv:1508.00496 and close keywords, the source paper, Orihuela’s 2016 conic James paper, and web searches on June 15, 2026 for the exact title, arXiv id, “Theorem 2”, “arbitrary Banach spaces”, and counterexample variants. The 2016 paper is the relevant later direct hit: it solves Theorem 10 and explicitly leaves Theorem 2 open. I found no later exact resolution in this bounded search.

Human Review Recommendation

Recommended review focus:

1. verify the equivalence in Theorem 1;
2. check the affine-slice shift argument in Lemma 1;
3. check that Theorem 3 is exactly an application of Orihuela’s conic theorem;
4. check the two strict inequalities in the frustum proof, since they are the mechanism forcing a maximizer into the far slice.
5. check the product calculation in Theorem 5, which is the mechanism forcing a maximizer to reveal the S -coordinate.

References

- [1] B. Cascales, J. Orihuela, and A. Pérez, *One-side James’ Compactness Theorem*, arXiv:1508.00496, 2015.
- [2] J. Orihuela, *Conic James’ Compactness Theorem*, arXiv:1610.02763, 2016.
- [3] R. C. James, *Weakly compact sets*, Trans. Amer. Math. Soc. 113 (1964), 129–140.